Production document

Audio Design Document

How the game sounds: principles, the mix, ambience, footsteps, spirits, music, voice, the full Episode 1 sound list and technical specifications.

Applies to	All episodes · Episode 1 sound list in full
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01 Audio Pillars

Pillar	In practice
Sound tells the truth	In the dark, sound is the player's main sense. Every threat must be locatable by ear alone, so spirit sounds are always spatial, distinct and honest (no fake positions).
Silence is a sound	Quiet stretches are planned. Insects and rain stopping is one of the strongest warnings in the game.
The world is alive	Rain, frogs, wood and wind are dense, layered and specific to Cambodia, never generic "horror ambience".
Music is rare	Most of the game has no music. When the Pinpeat tones arrive, they mean something.
Respect sacred sound	Chanting, bells and Pinpeat are used with permission and correct context, never distorted for cheap scares (section 6.3).

02 The Mix

Bus	Level guide	Rules
Voice	Loudest	Always intelligible; ducks ambience by 6 dB and music by 4 dB while speaking.
Threat	High	Never ducked. HRTF spatialisation for precise positioning.
SFX	Medium	Player's own sounds slightly quieter than threats so threats stand out.
Ambience	Low–medium	Two to four layers per zone; crossfade over 2–3 s between zones.
Music	Low	Sits under everything; never masks a threat.
UI	Low	Paper, pencil, satchel sounds; dry, non-spatial.

- **Loudness target:** the overall mix averages around −18 LUFS on a test playthrough, with peaks below −1 dBTP. Dialogue sits around −20 to −18 LUFS.
- **Dynamic range:** wide on purpose; horror needs quiet. A "night mode" setting compresses the range for players without headphones.
- **Headphones recommended:** the start screen says so; "headphone mode" sharpens 3D positioning.

03 Ambience

Zone (Ep 1)	Layers	Reverb
Studio (prologue)	Rain on glass · distant city traffic and a tram bell · room tone · clock ticking	Small room
Village path	Heavy rain on banana leaves · wind gusts · frogs · distant thunder	Open, dense foliage
Courtyard and tent	Rain on wet cloth · dripping from the tent edges · cicadas between showers · bullfrogs · the hiss of oil lamps	Open courtyard
Upper house	Rain on roof tiles · creaking teak · wind through shutters · muffled tent sounds through the floor gaps	Small wooden room
Under the house	Rain falling around the edges · dripping between floorboards · muddy ground · insects close up · floor creaks above	Low, semi-open

THE SILENCE RULE

Before every major threat, the insect and frog layers fade out over 3–5 seconds. Players learn quickly that silence means danger, and the game can use that expectation later.

04 Footsteps and Foley

4.1 Footstep matrix

Surface	Walk	Crouch	Hurry	Noise (0–1)
Teak floor	8 variations	6	8	0.5 · creaks
Dirt	8	6	8	0.3
Mud	8 (suction)	6	8	0.6
Dry leaves (Ep 2)	8	6	8	0.9
Stone	8	6	8	0.5
Moss (Ep 2)	6	6	6	0.15
Shallow water (Ep 3)	8 (splash)	6	8	0.7

Noise values feed the AI (Technical Design Document, section 10). Crouching halves them; hurrying doubles them.

4.2 Player foley

Group	Sounds
Clothing	Cotton blouse rustle on turning and crouching; heavier when wet
Satchel	Leather open, close, buckle; items shifting inside
Sketchbook	Page turns, pencil scratch for new entries, ink pen
Lamp	Wick turn, glass chimney lift, flame catch, flicker, sputter, shutter open/close, oil pour
Matches	Box slide, strike, flare, burn, blow out (several takes; failed strikes too)
Breathing	Calm, nervous, frightened, holding, gasp, shivering

05 Spirit Sound Design

Spirit	Signature	Source ideas
Neak Ta	Grinding stone under a deep, slow breath; a layered voice (low human, stone, distant wind); brass resonance when it is close.	Stones dragged on concrete, slowed; a brass bowl struck and bowed; voice layered and pitched down
Neak Ta shadow forms (Ep 1, 4, 5)	Wet, dragging footsteps; heavy breathing with a rattle; wood groaning under weight.	Wet cloth dragged on wood; breathing through cupped hands
Pret (Ep 2)	Thin, high, raspy breathing through a tiny mouth; long sniffing; a shriek when food is wasted.	Breathing through a straw; pitched and filtered
Pich (Ep 3, 5)	Light bare footsteps on dry boards, a soft giggle, a small wooden toy rolling; always from somewhere safe. Never processed to sound scary.	A child's footsteps on a wooden floor (recorded with consent and a parent present); a wooden toy
Chhlong Tonle (Ep 3)	Water moving where nothing moves it; muffled crying under water; fragments of a lullaby.	Hydrophone recordings; a voice recorded through water
Khmaoch Prey (Ep 4)	Perfect imitations of family voices, slightly wrong in timing and breath; creaking like wood when they move.	The real voice actors' lines, re-edited and lightly processed

Rule: each spirit has a sound the player can learn and that always means the same thing. Never play a spirit's signature unless that spirit is really there.

06 Music Brief

6.1 Direction

A sparse score built on traditional Khmer instruments, especially the **roneat** (រនាត) and the **Pinpeat** (ពិណពាទ្យ) ensemble, played slowly and quietly, with long silences, low drones and natural room sound. It should feel like music heard from far away across water at night.

6.2 Episode 1 cues

Cue	Moment	Description	Length
M1-01	Main menu	Solo roneat, slow, a simple melancholy phrase with long gaps; rain underneath	2–3 min loop
M1-02	Arrival (A1-02)	One low sustained tone and a few struck notes as the tent comes into view	30 s
M1-03	Portrait (A2-09)	A single dissonant struck chord, then silence	5 s
M1-04	Ritual (A3-07)	Low drone rising under the chant; no melody	40 s
M1-05	Dawn (R01)	Gentle roneat phrase (the menu theme, warmer)	45 s
M1-06	The name (R03)	Silence, then one quiet note as the blood spreads	15 s
M1-07	Credits	Full version of the main theme	2 min

6.3 Working with musicians

- Commission Cambodian musicians to record, or license existing recordings with clear written permission. Credit them by name.
- Ask the musicians which pieces and modes are appropriate for funerals and which must not be used in a game.
- Record instruments dry (close microphones) and with room sound, so they can be placed in the game's spaces.
- Do not use real chanting or ceremonial music as a horror sting, and do not reverse or distort it.

07 Voice Production

Language	Khmer for all characters (Episode 1 Script, section 1); English and Khmer subtitles.
Casting	See the Episode 1 Script, section 2. Prefer actors from or familiar with Battambang for Yay Mey's accent.
Recording	Quiet, treated room; large-diaphragm or broadcast dynamic microphone; 48 kHz, 24-bit WAV; 20–30 cm from the mouth; pop filter.
Takes	Three versions of each line (normal, quieter, more emotional) plus breaths and reactions (gasps, sighs) for Sora.
Processing	Light noise reduction, gentle compression, high-pass at 80 Hz. Sok's recording gets a gramophone treatment (band-limited, crackle, slight wow and flutter).
Files	`vo_<lineID>_t<take>.wav`, e.g. `vo_E1-A1-03-2_t2.wav`.

08 Episode 1 Sound List

Source: **R** record yourself · **L** library (check licence) · **D** designed from layers · **M** musicians. Priority: A = needed for the greybox, B = needed for release.

ID	Sound	Type	Source	Pri
S1-001	Rain layers: on glass, leaves, cloth, roof tiles, ground edges	Ambience loop	L + R	A
S1-002	Thunder (near, far) ×8	One-shot	L	A
S1-003	Cicadas, bullfrogs, night insects	Ambience loop	L	A
S1-004	Phnom Penh street at night 1932 (traffic, tram bell)	Ambience loop	L + D	B
S1-005	Wall clock tick and stop; clock chime (quarter hour)	Loop / one-shot	R	A
S1-006	Brass telephone ring; lifting the earpiece; static; dead line	One-shot / loop	L + D	A
S1-007	Phone scream, glass shatter, thuds (through the line)	One-shot	D	A
S1-008	Ink spill, pen scratch, candle blown out	One-shot	R	B
S1-009	Train and horse-carriage montage under the fade	One-shot	L	B
S1-010	Carriage stop, horse snort, wheels in mud	One-shot	L	B
S1-011	Footsteps: teak, dirt, mud (walk, crouch, hurry)	One-shot sets	R	A
S1-012	Teak creaks and groans (random, under weight)	One-shot set	R	A
S1-013	Lamp set (wick, chimney, catch, flicker, sputter, oil pour)	One-shot / loop	R	A
S1-014	Match set (box, strike, fail, flare, burn)	One-shot	R	A
S1-015	Satchel and sketchbook set	One-shot	R	A
S1-016	Hanging oil lamps hiss; wax candle flicker	Loop	R	B
S1-017	Incense crackle; burn-out puff	Loop / one-shot	R	B
S1-018	Whispers (incense neglected); layered, unintelligible	Loop	D	A
S1-019	Cold gust; lamps going out in a ripple (A2-01)	One-shot	D	A
S1-020	Wet footsteps above (A2-01)	One-shot	R + D	A
S1-021	Doors: wooden door open, creak, ajar swing; drawer open; lid slides	One-shot set	R	A
S1-022	Loose wall planks pulled away	One-shot	R	A
S1-023	Gramophone: crank, needle drop, crackle, speed dial, warped and corrected playback	Interactive	R + D	A
S1-024	Backward funeral chant on the gramophone	Loop	M + D	B
S1-025	Neak Ta shadow in the house: dragging steps, breathing, pause at door	Spatial set	R + D	A
S1-026	Wardrobe hide: door close, muffled filter, heartbeat	One-shot / loop	R + L	A
S1-027	Portrait reveal sound cue (subtle frame creak, intake of breath)	One-shot	D	B

ID	Sound	Type	Source	Pri
S1-028	Thread snap (sharp, echoing)	One-shot	R + D	A
S1-029	Corpse shift (cloth, wood creak), five variations	One-shot	R	A
S1-030	Clay jar lid lift; hand in water; fragment found	One-shot	R	A
S1-031	Neak Ta shadow under the house: patrol drag, wet breath, investigate sniff, chase rush	Spatial set	R + D	A
S1-032	Key fragments click together; chest lock and heavy lid	One-shot	R	A
S1-033	Presence behind Sora: breath at the ear, stone grind, the whispered name	Spatial, close	D	A
S1-034	Token burning; wind blast tearing the tent; fading screech	One-shot	D	A
S1-035	Dawn: birdsong, dripping, distant rooster	Ambience	L	B
S1-036	Blood spreading on paper (wet, slow)	One-shot	R + D	B
S1-037	Caught: scare hit, page tear, fade	One-shot	D	A
S1-038	Funeral chanting for the ritual	Loop	M (with permission)	B

09 Recording Plan

Session	What to record	Where / props
Wood session	Footsteps on old boards, creaks, doors, drawers, planks, chest, wardrobe	An old wooden house or floor; a wooden box
Fire session	Matches, a real hurricane lamp, candles, incense	Safe outdoor space; fire precautions
Paper and leather	Sketchbook, pages, pencils, fountain pen, satchel, letters	Quiet room
Water and mud	Hands in a clay jar, mud footsteps, dripping, wet cloth drags	Garden; a large pot of water
Rain (field)	Rain on leaves, cloth, tin and tiles in Cambodia during the monsoon if possible	Portable recorder, wind protection
Body	Breathing set, cloth movement, heartbeat	Quiet room

Equipment: a portable recorder (for example a handheld stereo recorder) plus one dynamic and one condenser microphone is enough. Record at 48 kHz / 24-bit, several takes, with a few seconds of room tone each time.

10 Technical Specifications

Item	Specification
Source files	48 kHz, 24-bit WAV in `art-source/audio/` (Git LFS)
Shipped format	Ogg Opus (96 kbps stereo ambience and music, 64 kbps mono effects and voice), with AAC fallback
Channels	Spatial effects and voice mono; ambience and music stereo
Loops	Seamless loop points; at least 30 s for ambience to avoid audible repetition
Streaming	Ambience and music streamed; short effects preloaded per area
Naming	`<bus>_<ep>_<name>_<variant>`, e.g. `sfx_ep1_teak_step_03.ogg`, `amb_ep1_tent_rain.ogg`, `mus_M1-04_ritual.ogg`
Budget	About 15–20 MB of audio in Episode 1's initial download; the rest streamed
Licences	Every library sound logged in the asset tracker with source, licence and credit